3. Filter it
4. Boil milk
5. Mix sugar and filtered coffee along with boiled milk
6. Pour the coffee into the cup to serve

This kind of procedure can be represented using an algorithm. Thus, the algorithm consists of step-step-by instructions that are required to accomplish a task and helps the programmer to develop the program.

Problem: Design an algorithm to find square of the given number and display the result.

The algorithm can be written as:

Step 1 – start the process

Step 2 – get the input x

Step 3 –calculate the square by multiplying the input value ie., square ← x* x

Step 4 – display the result square

Step 5 – stop

Algorithm could be designed to get a solution of a given problem. A problem can be solved in many ways. Among many algorithms the optimistic one can be taken for implementation.

An algorithm that yields expected output for a valid input is called an algorithmic solution.

Algorithm	Program
• Algorithm helps to solve a given problem logically and it can be contrasted with the program	• Program is an expression of algorithm in a programming language
• Algorithm can be categorized based on their implementation methods, design techniques etc	• Algorithm can be implemented by structured or object oriented programming approach
• There is no specific rules for algorithm writing but some guidelines should be followed.	• Program should be written for the selected language with specific syntax
• Algorithm resembles a pseudo code which can be implemented in any language	• Program is more specific to a programming language

Table 4.1 Algorithm Vs Program

4.1.3. Analysis of Algorithm

Computer resources are limited. Efficiency of an algorithm is defined by the utilization of time and space complexity.